

JAINAM SHAH

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EDUCATION

New Jersey Institute of Technology, Albert Dorman Honors College

May 2027

Bachelor of Science in Computer Science, Minor in Mathematics + Business

Newark, NJ

- GPA: 3.93 | Honors Merit Scholarship (Full Tuition)
- Relevant coursework: Data Structures & Algorithms, Compiler Design, Generative AI, Machine Learning, Database Management, Linux and C programming, Network Security, Operating Systems, Linear Algebra

SKILLS & CERTIFICATIONS

Softwares: Anaconda, WSL, LaTeX, Xampp, Minitab, Git, Visual Studios, Postman, Jira, Confluence, Jupyter Notebooks, Docker

Technical Skills: Python, Java(Spring Boot), C++, HTML, CSS, JavaScript, SQL, Linux, Agile software development, Scrum framework

EXPERIENCE

TD Securities - Risk Management and Product Control

June 2025 – Present

Software Engineering Intern

New York City, NY

- Reduced **14K** risk mappings to **~100 regex-based patterns** through data analysis, streamlining risk attributions across TD systems.
- Leveraged **Java Spring Boot** to monitor **80+** critical APIs over **REST and ZeroMQ**, with a **real-time UI** for actionable insights.
- Parallelized API checks and trade metric validation (NPV, PnL) using a **multithreaded pipeline**, reducing execution time by **~70%**.
- Employed **Protobuf** to decrease latency by **52%** and streamline cross-platform communication across **4+ backend services**.

Office of Naval Research / Big Data Analytics Laboratory

Feb 2024 – Dec 2024

Machine Learning Intern / Honors Research Fellow & Grant Recipient

Newark, NJ

- Utilized Machine Learning skills to help the US NAVY predict ship maintenance delays and optimize resource allocation.
- Enhanced model accuracy by **80%**, reducing predictive errors to **14 days** and saving the U.S. Navy over **\$1 billion** in costs.
- Analyzed over **15k** classified features, providing data-driven recommendations on model functionality for U.S. Navy officials.
- Reduced overfitting by **20%**, while increasing accuracy by **10%** through a custom algorithm harmonizing ML output.

Tech for Good Inc.

July 2024 – Sept 2024

Software Engineering Intern

Boston, MA

- Engineered an **openEHR-compliant EMR system** in **Go** to digitize rural patients' records, initially deployed in Tanzanian Hospitals.
- Designed and implemented a **ScyllaDB** schema with **5000+** records, exposed via performant **GraphQL API endpoints**.
- Implemented **Docker** to ensure consistent database and language schemas across local machines for all team members.

Structural Analysis of Biomedical Ontologies Center (SABOC)

Jan 2024 – May 2024

Machine Learning Intern

Newark, NJ

- Extract combinations of over **37 million** citations from the PubMed database to identify different **Social Determinants of Health**.
- Employed **NumPy** and **Pandas** to filter text file redundancies and create a unique list SDoH, increasing the ontology by **15%**.
- Integrated **UMLS API** to extract **20k+** synonyms for subject/predicate pairs, expanding the ontology with semantic equivalents.

PROJECTS

[Simple Fortran 95-like compiler](#) - C++

- Built a compiler (**lexical analyzer**, **parser**, and **interpreter**) to read and understand a Simple Fortran 95-like language

[Stock-Market Analyzer](#) - Django, HTML, Twelvedata-API, SQLite

- Built a stock portfolio dashboard with live equity pricing, portfolio valuation from simulated trades, and user-specific preferences

[WasteFree](#) - Kolang, Python, SQLite, Mindee-API, Beautiful Soup, Bcrypt

- Built a **Kolang** system to track pantry inventory, send expiration alerts, and suggest user-endorsed recipes to reduce food waste.
- Integrated **Mindee API** to extract items from grocery receipts and update a normalized **SQLite database** for pantry updates.

[GrooveSync](#) - React.js, Express.js, Node.js, MongoDB, Spotify API, Anthropic Claude Ai API

- Built a disco app using **React**, **Node**, **Express**, and **MongoDB**, integrating **Spotify API** for music playback and interactive features.
- Developed **DiscoAI** and a **Claude**-powered narrative game to offer dynamic insights into disco culture and history.

LEADERSHIP EXPERIENCE & AFFILIATIONS

CodePath - Community Member & Student

Feb 2025 - Present

NJIT Investment Fund - Junior Analyst

Dec 2023 - Jan 2025

Bergen County Debate Club - Coach & Alum

Sept 2021 - Present